
ScoreInfo APIs – Specifications

Version 1.9 – May 2022

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Version No.	Date	Conqueror Version	Description
1.9	May 2022	14.22.00	Added GetReservationScores call
1.8.2	May 2022		Added PinsDown field
1.8.1	March 2022		Added CDEIDSquad field
1.8	February 2022		Added CDEReference
1.7	December 2017	12.14.00	Added field to detect the game completion
1.6	June 2017	12.06.00	Added fields related to Conqueror tournament competitions
1.5	March 2017		Added reference to the 'Internet Gateway and Conqueror Pro API - Cloud Connection' document
1.4	February 2017	11.32.01	APIs available through QubicaAMF Cloud Connection
1.3	June 2015		Added reference to the Internet Gateway document
1.2	17 th September 2013		Added field PlayerFullName in the Game XML entity
1.1	5 th September 2013		Added field IDEvent in the Event XML entity Added field IDGame in the Game XML entity
1.0	23 rd February 2012	9.10.00	Specifications of all APIs

Index

1	Introduction.....	3
2	Prerequisites.....	3
3	Access to the APIs directly on the Conqueror Server	4
3.1	HMAC Authentication (Revision A).....	4
4	Access to the APIs using the QubicaAMF Cloud Connection.....	5
4.1	HMAC Authentication (Revision B).....	5
5	APIs.....	6
6	Requests and responses.....	10
6.1	Scores.....	11
6.1.1	GetScores	11
6.1.2	GetLastScores.....	13
6.1.3	GetReservationScores.....	15
7	HTTP Status Codes.....	17

1 Introduction

The purpose of this document is to specify the resources and operations required as part of the RESTful web service built by QubicaAMF providing the information about played games.

The RESTful web service built by QubicaAMF will use the HTTPS protocol for secure transmission, XML for resource representation and HMAC for authentication.

There are two alternative methods of utilizing the ScoreInfo APIs:

- Access to the APIs directly on the Conqueror Server
- Access to the APIs using the QubicaAMF Cloud Connection

The second one is the recommended method.

2 Prerequisites

The 'Score Information API' licence is required.

If you access the APIs through the QubicaAMF Cloud Connection, a personal account must be issued by QubicaAMF to authenticate your external application. Furthermore, the "Cloud Connection" licence is required.

3 Access to the APIs directly on the Conqueror Server

In this case the APIs are accessible over HTTPS on port 10011 of the Conqueror server.

e.g.: `https://{CenterAddress}:10011/Scores`

Where *{CenterAddress}* is the 'public' IP address of the Conqueror server (for example the IP address of the server in your VPN).

For the bowling center IT requirements, please refer to the "Internet Gateway and Conqueror Pro API – Direct Access" document.

3.1 HMAC Authentication (Revision A)

The access to the APIs is granted only to authenticated clients.

Refer to the "HMAC Authentication for Conqueror Pro API - RevA" document for implementation details and examples.

The application name and the associated secret key are fixed for this API version:

Application name: **Conqueror**

Secret key: **C7DD662A-8AAD-4EA0-BE7F-5CE43586E950**

4 Access to the APIs using the QubicaAMF Cloud Connection

In this case APIs are accessible over HTTPS on the internet domain ***centerapi.qubicaamf.com***

The template URL is:

https://centerapi.qubicaamf.com/rest/{CenterID}/ScoreInfo/{Resource}

where *{CenterID}* is the Qupee Center ID associated to the center.

e.g.: *https://centerapi.qubicaamf.com/rest/1234/ScoreInfo/Scores*

For the bowling center IT requirements, please refer to the “Internet Gateway and Conqueror Pro API - Cloud Connection” document.

4.1 HMAC Authentication (Revision B)

The access to the APIs is granted only to authenticated clients.

Refer to the “HMAC Authentication for Conqueror Pro API - RevB” document for implementation details and examples.

The application name and the associated secret key are personal and must be issued by QubicaAMF.

5 APIs

This section describes the resources, including how resource states are addressed and transferred over HTTPS.

The XML string transmitted is encoded using UTF-8.

HTTP Method	URI Template	Operation
GET	/Version	GetAPIVersion Returns the version of the APIs
GET	/Scores?lanes={lanes}&fromDate={fromDate}&toDate={toDate}	GetScores Reads all the scores from the lanes. Request data are the following: • lanes (optional) Indicates the lanes which scores have to be read. Lanes are separated by a comma to select single lanes (e.g.: lanes=1,3,4 selects lanes 1, 3 and 4) or a dash to select a laneset (e.g. lanes=1-4 is equivalent to lanes=1,2,3,4) If no lane set is specified, the scores of all the lanes will be returned. • fromDate (optional) Indicates the date from which the scores have to be returned (YYYYMMDD). This parameter is optional. If no <i>fromDate</i> is specified, the scores of the last event will be returned and the <i>toDate</i> parameter will be ignored. • toDate (optional) Indicates the date and time to which the scores have to be returned (YYMMDD). The scores are returned into the Score data.
GET	/Scores/Last/{lanes}	GetLastScores Reads all the last scores from the lanes (comma separated, e.g.: 1,2,5,6). The scores are returned into the Score data.
GET	/Scores/Reservation/{idReservation}	GetReservationScores Reads all the scores of the reservation. The scores are returned into the Score data.

<i>Scores</i>	
Field	Description
Events	List of the events (Event).

<i>Event</i>	
Field	Description
IDEvent	Unique identifier for the Event.
IDReservation	Unique identifier for the Reservation (only for GetReservationScores call)
IDTournament	Unique identifier for the Tournament competition.
IDTournamentEvent	Unique identifier for the Tournament Event competition.
IDSquad	Unique identifier for the Squad competition.
CDEIDSquad	External unique identifier for the CDE Squad competition.
OpenDateTime	The date and time when the event started (YYYYMMDDHHMM).
CloseDateTime	The date and time when the event finished (YYYYMMDDHHMM).
Status	Current status of the event (NotPlayed , Playing , Played)
OpenMode	Indicates the lane opening mode (Single , Pair , Group , League , Tournament)
Games	List of the games (Game) played by each player.
Bonuses	List of the bonuses (Bonus) earned by each team in the competition.

<i>Game</i>	
Field	Description
IDGame	Unique identifier for the Game.
IDPlayerCompetition	Unique identifier for the player in the competition
CDEReference	External CDE Reference (CDEReference).
IDTeamCompetition	Unique identifier for the team in the competition
LaneNumber	Number of the lane.
PlayerName	Short name (nick name) of the player (e.g. "Jim").
PlayerFullName	Full name of the player (e.g. "Jim Morrison").
FreeEntryCode	An alphanumeric code that can be used for registering the Federation sanction ID.
PlayerPosition	Ordinal position of the player.
StartDateTime	The date and time when the game started (YYYYMMDDHHMM).
EndDateTime	The date and time when the game terminated (YYYYMMDDHHMM).
Completed	Yes in case of game completed, otherwise No .
GameNumber	Number of the game.

Hdcp	Player handicap.
Total	Total with handicap.
Frames	List of the played frames (Frame).

<i>Frame</i>	
Field	Description
FrameNumber	Progressive number of the frame.
Bowl1	Info related to the first throw (Bowl).
Bowl2	Info related to the second throw (Bowl).
Bowl3	Info related to the third throw (Bowl - 12 th frame only).
ProgressiveTotal	Total score since the beginning of the game.
IsConvertedSplit	Yes in case of converted split, otherwise No .

<i>Bowl</i>	
Field	Description
Total	Total pins knocked down in the analysed frame.
IsStrike	Yes in case of strike, otherwise No .
IsSpare	Yes in case of spare, otherwise No .
IsSplit	Yes in case of split, otherwise No .
IsGutter	Yes in case of gutter, otherwise No .
IsFoul	Yes in case of foul, otherwise No .
IsManuallyModified	Yes if the score has been manually modified, otherwise No .
Pins	A comma separated list of the knocked down pins (e.g.: 1,2,3,4,5,6,7,8).
PinsDown	Count of knocked down pins of the analysed throw.

<i>Bonus</i>	
Field	Description
IDTeamCompetition	Unique identifier for the team in the competition.
SerieNumber	Number of the serie.
GameNumber	Number of the game.
Points	Points earned.

<i>CDE</i> Reference	
Field	Description
IDSquad	Unique identifier for the squad in the competition.
IDRegistration	Unique identifier for the registration.
IDPlayer	Unique identifier for the player.

6 Requests and responses

In this section are specified the xml messages sent and received in the HTTP body, for the various resources and operations.

We adopted the following conventions:

- The lane URLs contained in the xml messages are relative
Example for lane 5: ***/Scores?lanes=5&fromDate=20111128=&toDate=20111128***
- The *date fields* format is **YYYYMMDD**
Example for 31 January 2012: ***20120131***
- The *time fields* format is **HHMM**
Example for 18:45: ***1845***
- The Boolean fields are represented by values: ***Yes*** and ***No***

6.1 Scores

6.1.1 GetScores

URI Template

GET /Scores?lanes=1&fromDate=20111128=&toDate=20111128

Response

```
<?xml version="1.0" encoding="utf-8"?>
<Scores xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns:xsd="http://www.w3.org/2001/XMLSchema">
  <Events>
    <Event>
      <IDEvent>62</IDEvent>
      <IDTournament>5</IDTournament>
      <IDTournamentEvent>8</IDTournamentEvent>
      <IDSquad>2</IDSquad>
      <OpenDateTime>201111281654</OpenDateTime>
      <Status>Playing</Status>
      <OpenMode>Single</OpenMode>
      <Games>
        <Game>
          <IDGame>68</IDGame>
          <IDPlayerCompetition>1</IDPlayerCompetition>
          <IDTeamCompetition>1</IDTeamCompetition>
          <LaneNumber>1</LaneNumber>
          <PlayerName>James</PlayerName>
          <PlayerFullName>James Brown</PlayerFullName>
          <FreeEntryCode>10-306468</FreeEntryCode>
          <PlayerPosition>1</PlayerPosition>
          <StartDateTime>201111281655</StartDateTime>
          <EndDateTime>201111281656</EndDateTime>
          <Completed>No</Completed>
          <SerieNumber>1</SerieNumber>
          <GameNumber>1</GameNumber>
          <Hdcp>0</Hdcp>
          <Total>35</Total>
          <Frames>
            <Frame>
              <FrameNumber>1</FrameNumber>
              <Bowl1>
                <Total>10</Total>
                <IsStrike>Yes</IsStrike>
                <IsSpare>No</IsSpare>
                <IsSplit>No</IsSplit>
                <IsGutter>No</IsGutter>
                <IsFoul>No</IsFoul>
                <IsManuallyModified>No</IsManuallyModified>
                <Pins>1,2,3,4,5,6,7,8,9,10</Pins>
                <PinsDown>10</PinsDown>
              </Bowl1>
              <ProgressiveTotal>18</ProgressiveTotal>
              <IsConvertedSplit>No</IsConvertedSplit>
            </Frame>
          </Frames>
        </Game>
      </Games>
      <Bonuses>
        <Bonus>
          <IDTeamCompetition>1</IDTeamCompetition>
          <SerieNumber>1</SerieNumber>
          <GameNumber>1</GameNumber>
          <Points>1</Points>
        </Bonus>
      </Bonuses>
    </Event>
  </Events>
</Scores>
```

Response if request is not valid

Status code: 400

Response in case of internal server error

Status code: 500

6.1.2 GetLastScores

URI Template

GET /Scores/Last/1

Response

```
<?xml version="1.0" encoding="utf-8"?>
<Score xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns:xsd="http://www.w3.org/2001/XMLSchema">
  <Events>
    <Event>
      <IDEvent>62</IDEvent>
      <IDTournament>5</IDTournament>
      <IDTournamentEvent>8</IDTournamentEvent>
      <IDSquad>2</IDSquad>
      <OpenDateTime>201111281654</OpenDateTime>
      <Status>Playing</Status>
      <OpenMode>Single</OpenMode>
      <Games>
        <Game>
          <IDGame>69</IDGame>
          <IDPlayerCompetition>1</IDPlayerCompetition>
          <IDTeamCompetition>2</IDTeamCompetition>
          <LaneNumber>1</LaneNumber>
          <PlayerName>James</PlayerName>
          <PlayerFullName>James Brown</PlayerFullName>
          <FreeEntryCode>10-306468</FreeEntryCode>
          <PlayerPosition>1</PlayerPosition>
          <StartDateTime>201111281655</StartDateTime>
          <EndDateTime>201111281656</EndDateTime>
          <Completed>No</Completed>
          <SerieNumber>1</SerieNumber>
          <GameNumber>1</GameNumber>
          <Hdcp>0</Hdcp>
          <Total>35</Total>
          <Frames>
            <Frame>
              <FrameNumber>1</FrameNumber>
              <Bowl1>
                <Total>8</Total>
                <IsStrike>No</IsStrike>
                <IsSpare>No</IsSpare>
                <IsSplit>No</IsSplit>
                <IsGutter>No</IsGutter>
                <IsFoul>No</IsFoul>
                <IsManuallyModified>No</IsManuallyModified>
                <Pins>1,2,3,4,5,6,7,8</Pins>
                <PinsDown>8</PinsDown>
              </Bowl1>
              <ProgressiveTotal>8</ProgressiveTotal>
              <IsConvertedSplit>No</IsConvertedSplit>
            </Frame>
          </Frames>
        </Game>
      </Games>
      <Bonuses>
        <Bonus>
          <IDTeamCompetition>2</IDTeamCompetition>
          <SerieNumber>1</SerieNumber>
          <GameNumber>1</GameNumber>
          <Points>1</Points>
        </Bonus>
      </Bonuses>
    </Event>
  </Events>
</Score>
```

Response if request is not valid

Status code: 400

Response in case of internal server error

Status code: 500

6.1.3 GetReservationScores

URI Template

GET /Scores/Reservation/1237

Response

```
<?xml version="1.0" encoding="utf-8"?>
<Score xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns:xsd="http://www.w3.org/2001/XMLSchema">
  <Events>
    <Event>
      <IDEvent>62</IDEvent>
      <IDReservation>1237</IDReservation>
      <IDTournament>5</IDTournament>
      <IDTournamentEvent>8</IDTournamentEvent>
      <IDSquad>2</IDSquad>
      <OpenDateTime>201111281654</OpenDateTime>
      <Status>Playing</Status>
      <OpenMode>Single</OpenMode>
      <Games>
        <Game>
          <IDGame>69</IDGame>
          <IDPlayerCompetition>1</IDPlayerCompetition>
          <IDTeamCompetition>2</IDTeamCompetition>
          <LaneNumber>1</LaneNumber>
          <PlayerName>James</PlayerName>
          <PlayerFullName>James Brown</PlayerFullName>
          <FreeEntryCode>10-306468</FreeEntryCode>
          <PlayerPosition>1</PlayerPosition>
          <StartDateTime>201111281655</StartDateTime>
          <EndDateTime>201111281656</EndDateTime>
          <Completed>No</Completed>
          <SerieNumber>1</SerieNumber>
          <GameNumber>1</GameNumber>
          <Hdcp>0</Hdcp>
          <Total>35</Total>
          <Frames>
            <Frame>
              <FrameNumber>1</FrameNumber>
              <Bowl1>
                <Total>8</Total>
                <IsStrike>No</IsStrike>
                <IsSpare>No</IsSpare>
                <IsSplit>No</IsSplit>
                <IsGutter>No</IsGutter>
                <IsFoul>No</IsFoul>
                <IsManuallyModified>No</IsManuallyModified>
                <Pins>1,2,3,4,5,6,7,8</Pins>
                <PinsDown>8</PinsDown>
              </Bowl1>
              <ProgressiveTotal>8</ProgressiveTotal>
              <IsConvertedSplit>No</IsConvertedSplit>
            </Frame>
          </Frames>
        </Game>
      </Games>
      <Bonuses>
        <Bonus>
          <IDTeamCompetition>2</IDTeamCompetition>
          <SerieNumber>1</SerieNumber>
          <GameNumber>1</GameNumber>
          <Points>1</Points>
        </Bonus>
      </Bonuses>
    </Event>
  </Events>
</Score>
```

Response if request is not valid

Status code: 400

Response in case of internal server error

Status code: 500

7 HTTP Status Codes

The following HTTP status codes will be used in the response:

HTTP Status Code	Description
200	Success. A request was fulfilled.
400	Bad request. Will be returned if any validation errors occur or received messages are incorrectly formatted.
401	Unauthorized. Will be returned if authentication fails.
404	Not Found. Will be returned if the server has not found anything matching the specified URI.
409	Conflict. The request could not be completed due to a conflict with the current state of the resource.
500	Internal Server Error. Will be returned in case of an unexpected error while processing the request.